Adaptive Regime-Switching Mean-Reversion on the NIFTY-50 Index

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Abstract

In this project, we design a regime-aware mean-reversion (MR) trading framework for the NIFTY-50 to enhance robustness, tail-risk control, and performance stability. It mixes market-dynamics diagnostics, fractal persistence metrics, and Hidden Markov Model–based volatility regimes, with regime-sensitive position sizing and layered risk controls. We analysed the NIFTY 50 data with different frequencies to generate signals that perform well in anti-persistent, low-trend environments, usually fail in trending, high-volatility, or in so-called event-driven regimes. The project, which we are developing, explores how fractal persistence measures, latent volatility regime models, and market-context conditioning can be integrated into a trading framework in a way that is adaptive. This framework will have several parts as follows: regime-sensitive risk allocation, as well as layered tail-risk controls. Strategy variants are evaluated net of transaction costs using walk-forward validation and controlled ablations to isolate causal contributions.

Keywords: Mean Reversion; Regime-Aware Trading; Volatility Regimes; Hidden Markov Models; Hurst Exponent; Fractal Persistence; Market Microstructure; Volatility Clustering; Heavy Tails; Stationarity; Tail Risk; Conditional Value-at-Risk (CVaR); Drawdown Convexity; Regime-Sensitive Position Sizing; Fractional Kelly; Volatility Targeting; Risk Management; Walk-Forward Validation; Ablation Analysis; Algorithmic Trading; Regime-switching dynamics; NIFTY-50

Introduction

This project builds and tests a regime-aware mean-reversion (MR) trading framework for the NIFTY-50 index (and liquid constituents), based on market microstructure diagnostics, persistence theory, latent volatility regimes, and layered risk controls. The core hypothesis is that mean-reversion strategies behave more robustly, less crash-prone, and are stabilised under various market settings if their signals, positioning, and risk limits are conditioned on

- Statistical properties of returns,
- Fractal persistence metrics,
- Latent volatility regimes [1][3][6].

The design is based on a modular, auditable pipeline. Every research module generates repeatable artefacts (notebooks, Python modules, CSV output, HTML reports, etc) which are passed along to the downstream components for walk-forward evaluation and ablation. All improvement claims are checked after

transaction costs and execution lags are factored in, and are tested for incremental contribution by controlled ablation [11][15].

This is an absolute return model which aim to maximize Sharpe Ratio and Kestner Ratio.

Problem Statement

The regime dependence of mean-reversion strategies in large-cap stocks of Indian equity indices is evident. To drive this point home a little more: signals that do perform well in anti-persistent, low-trend environments usually fail in trending, high-volatility, or in so-called event-driven regimes. Most existing implementations depend on fixed rules for signal thresholds, position sizing, and risk management. This causes significant performance deterioration, increased drawdowns, and poor tail-risk behaviour, particularly when market regimes change. The project, which we are developing, explores how fractal persistence measures, latent volatility regime models, and market-context conditioning can be integrated into a trading framework in a way that is adaptive. This framework will have several parts as follows: regime-sensitive risk allocation, as well as layered tail-risk controls. In order to do so, we strive to develop and validate one such theoretically informed methodology that extends the existing mean-reversion strategy for different market regimes. We hope that this will help us derive regime identifiers and signalling alpha, which we can use in trading engines and models.

Literature Review

1. Short-Term Reversals and Trading Activity

Typically, stock returns (stocks that do well for a week or month and then do poorly in the following week or month, and the reverse) are in fact short-term reversals (over the week or months) that occur regularly in financial markets. This was first formally noted by Jegadeesh (1990)[13] and Lehmann (1990)[17]. This realisation directly contradicts the stringent version of the Efficient Market Hypothesis (EMH), which assumes independence of future returns from past returns. The extent and continuance of these reversals have been studied on the basis of trading activity, yet so far, mixed results have emerged. Conrad et al. (1994)[7] demonstrated that weekly returns for stocks with increased trading activity display stronger reversals than those for stocks with low levels of trading activity. On the other end of the spectrum, Cooper (1999)[8] proposed that weekly returns of high trading volume stocks have a weaker reversal effect than those of low trading volume stocks. Avramov et al. (2006)[3] found that the monthly reversal effect of returns decreases with the turnover increase, and at weekly frequency, reversal effects are synonymous with higher trading activity. Taken together, these results emphasise the significance of frequency-dependence and traders' exchange activity in short-term return dynamics. David et al. (2012)[28] implemented a new reversal strategy using residual stock returns which do not display these exposures, and in so doing achieved twice as high risk-adjusted returns as those of a conventional reversal strategy. And they showed that those residual reversal strategies are effective and statistically and economically profitable even when accounting for trading costs. Chris et al. (2019)[6], by the trading activities, also resolve the interplay between reversal effects and momentum effects. Their research demonstrates that the stocks with the highest

turnover demonstrate short-term momentum rather than reversals. They analysed these results through a combination of effects. Short-term reversals may stem from short-term liquidity needs first. Second, news-driven returns tend to continue rather than reverse. Third, turnover serves a dual role in liquidity and news-driven trading activity. Thus, the evidence shows that momentum takes priority over reversal as turnover increases.

2. The Fractal Market Hypothesis (FMH) and the Hurst Exponent

As such, Peters (1994)[21] also developed the Fractal Market Hypothesis (FMH) as an alternative framework to the Efficient Market Hypothesis to account for nonlinear dynamics in asset prices and deviations from market efficiency. It is known to be founded on chaos theory and fractal geometry, where FMH considers the financial sector to be a set of heterogeneous investors who are trading across a period of time with varying investment horizons, causing periodic volatility. This framework was derived from the Hurst Exponent that was originally developed by Hurst (1951)[12]. This exponent was first formulated in hydrology by hydrogeologist Harold Edwin Hurst (1951). In his case, he was trying to find the optimal dam size of the Nile River for the volatile drought and rain scenarios. The Hurst exponent (H) ranges between 0 to 1. The time series is divided into three types: (1) $H = 0$ (random series) (2) $0 < H < 0.5$ — It indicates an anti-persistent or mean-reverting time series. As H approaches 0, the mean-reversion is stronger; (3) $0.5 < H < 1$ — It indicates that there is a persistent time series (or trending): hence, the direction of the next value is more likely to be the same as the current value. There has been a lot of work performed on the Hurst exponent in various financial markets to evaluate the productivity and long-memory properties of the market, as well as to evaluate asset type. Empirical studies have demonstrated its applicability to equity indices (Matos et al., 2008[18]), commodities (Kristoufek, 2019[15]; Tiwari et al., 2021[24]), fixed-income markets (Bariviera et al., 2012[4]), foreign exchange (Shahzad et al., 2018[22]), and cryptocurrency (Dimitrova et al., 2019[9]; Kristoufek & Vosvrda, 2019[16]) as well. When all of these studies are taken together, we can understand that the persistence of the financial markets is in opposition to constant information efficiency. Besides the stationary measures, other studies examined time-varying or local Hurst exponent estimates that identify structural changes and extreme market events (and other) and the importance of such estimates. Grech and Mazur (2004)[10] studied the 1929 and 1987 crashes and found Hurst exponent technique could give vital signals on extreme events. The same findings have been found in similar work by Czarnecki et al. (2008) and Grech and Pamuła (2008) for the Polish WIG20 index, and Morales et al. (2012)[19] generalised this approach to U.S. equity portfolios. They showed that different ranges of the Hurst exponent correspond to different phases of market evolution. They showed how sharp changes in the dynamical curve of the Hurst exponent usually precede structural breaks, and Nicole and Joseph (2013)[1] studied what fractal properties mean for financial markets. They modelled the mechanisms of investor heterogeneity and price formation, thereby connecting fractal dynamics to financial (in)stability.

3. Modelling Market Regimes Using Markov Switching Frameworks

Also, a new sub-family of models called the econometric regime-switching models offers an alternative model for how models would capture complex

nonlinear dynamics in financial time series. Fractal analysis is usually integrated with another one, such as the Markov Switching Model (MSM) or the Hidden Markov Model (HMM). MSM, which follows the influential theory of Hamilton (1989)[11], postulates that observed asset returns are the result of the latent states that evolve according to a Markov process. It is based on market dynamics, and market dynamics depend on regimes (bull vs. bear markets, for example). Contrary to linear models, MSM can accommodate 2 discrete shifts in mean and volatility, allowing for multiple structural breaks. They can thus capture asymmetries among different market regimes. Chan et al. (2017)[5] demonstrated the effectiveness of the MSMs in capturing nonlinear financial dynamics which result from the phase transitions of the underlying regime. Kristoufek (2013)[14] demonstrates empirically that financial crises take place within a single investment horizon and when market liquidity dries up, which is consistent with regime-dependent market behaviour. Market regime transitions, especially bull-versus-bear transitions, are significant in terms of practice for investors and risk managers (Wang, 2008[31]). MSMs and similar Hidden Markov Models (HMMs) have been widely used to discover such transitions. Wang et al. (2020)[25] found that HMMs are able to segment asset-return series into latent regimes with specific return and volatility traits. Adding regime information to trading strategies also enhances risk-adjusted performance. Similarly, Khaled et al. (2016)[23] showed that HMMs integrated with EVT improve Value-at-Risk forecasting in a high-volatility environment.

Competitor Analysis

As we outlined in the Literature review, Hurst exponent and Hidden Markov-based models have been widely used in financial market research to study the underlying dynamics of market behaviour. They are used as generalisations for the structure of memory and market persistence (price structure including trend-following vs. mean-reverting) as well as latent market states (volatility states and transitions between stable and unstable market conditions). Christos (2009)[20] presented a Duration Hidden Markov Model for different market regimes (e.g., bull and bear) switches in the stock market. In this model, the duration of every Markov Chain state is treated as a random variable in the analysis. On NYSE returns, they extended the model using short-term interest rate and interest rate spread variables. In the same way, Yuan and Gautam (2016)[26] applied a Hidden Markov Model method with the FTSE 100 and Euro Stoxx 50: they aimed to pick up stylised factors such as fat tails and volatility clustering. In their study on the topic, which challenged stochastic methods including the HMM, Piotr (2019)[29] presented a rule-based method to identify the Bull and Bear Market Regimes. He demonstrated the application of this method to Dow Jones Industrial Average index data. For volatile and dynamically-driven emerging markets, this approach is not appropriate, however. Assoe (1998)[2] found a use of Markov-switching models on nine emerging markets, and his findings strongly support regime-switching in emerging market returns. Mean–reversion strategies are commonly found in equity markets, but since their application varies greatly by market regime, so too do their results. In Sideways markets, low volatility or Anti-persistent price behavior typically performs very well. But they do poorly in strong trends, high volatility and event-driven or news-rich phases. This contradiction is particularly observed among the equity markets in India. Most of the mean-reversion strategies are, in fact, performing worse than expected because

static market dynamics are assumed, and the underlying laws, such as fixed z-score thresholds, constant position size, and consistent stop-loss rules, are used. Static designs like this do not consider the changes of price persistence, volatility regimes, and tail-risk dynamics, leading to weak robustness and extreme drawdowns in challenging market times. Our key research addresses this gap by suggesting a regime-aware, adaptive mean-reversion trading framework that conditions signal generation and risk allocation on change of the structure of markets, therefore better ensuring that the framework can withstand fluctuations and resist adverse effects across heterogeneous market regimes. To determine fractal persistence, we employed a rolling Hurst exponent calculated through rescaled range analysis and subsequently classified the resulting market regimes into anti-persistent state, neutral state, and persistent state to provide regime-aware signal conditioning in real time. The main objective of the focus in this work is the construction of a practical trading framework based on a statistical or structural perspective of moving prices and volatility regime shifts on short-to-medium term price behaviour. Our framework, unlike most solutions, will not be based on one static stop-loss mechanism, as is done in the majority of works, but will use multiple dynamic risk factors implemented to enhance drawdown losses and tail risk behaviour to keep up with the changing market regimes. Our work is mainly carried out with respect to Indian equity indices and large-cap stocks.

Theoretical Foundation

1. Market Dynamics

As we know, market data is non-stationary and available to various time intervals. Most of the models work with stationary data. There are various methods (ADF, KPSS) to test the stationarity. Since mostly the market data is non-stationary, there are standard techniques like first or second order differencing to transform non-stationary data into stationary data. There are also scenarios to analyse and diagnose the volatility. We used the Ljung-Box on realised volatility to examine volatility clustering. The Gaussian distribution is often used to model the market. It underestimates the risk on the tail. We used the t-distribution and the Cornish–Fisher Expansion to analyse fat tails. The p-values and deviations from Gaussian assumptions of data are calibrated with Monte-Carlo white-noise benchmarks. This identifies the failure points of classical MR assumptions and grounds regime conditioning empirically

2. Hurst Exponent

Hurst Exponent is one of the popular techniques to analyse the long-term dependency of time series. It has many variants like R/S, DFA and wavelet. The R/S estimator is calculated as the range of cumulative deviations from the mean in each sub-sample, normalised by the standard deviation of the sub-sample and related to the sub-sample size Ω and empirically shown as:

$$H = \ln(R/S) / \ln(\Omega)$$

In the presence of memory, Hurst Exponent(H) exhibits persistence and its value of greater than 0.55. The value of H less than 0.45 signifies anti-persistent behaviour. The value between 0.45 and 0.55 signifies random behaviour.

3. Hidden Markov Model (HMM)
The Hidden Markov Model is based on the Markov chain. A Markov chain is a probabilistic graphical representation of a stochastic process that transitions between a finite number of states. It is based on the assumption that the future state only depends on the current state.

$$P(q_i = a|q_1 ... q_{i-1}) = P(q_i = a|q_{i-1})$$

The HMM is a Markov chain-based probabilistic model where the sequence of observable variables is generated by a sequence of hidden states. The hidden states are not observed directly. For example, the observed variables can be like price, while the hidden states can be a market regime. The HMM is used to detect the market regime, including upward trends, downward movements, and sideways markets.

4. Anatomy of Mean-Reversion Signal
We use a vectorised horizon grid of z-score lookbacks and exit rules to construct a baseline MR strategy [13]. Signal decay and long-short asymmetry are quantified by means of an impulse-response curve (PnL vs holding period). Execution-lag sensitivity at 1, 2, and 5 ticks, to test their susceptibility to realistic microstructure effects [14].

5. Regime-Sensitive Sizing of Positions
Policies for position sizing are then considered as a function of regime probability and volatility, vol-target, fractional-Kelly, and hybrid regimes [10]. Policies are compared based on a drawdown-convexity indicator (ΔMaxDD / Δσ_exposure) and tail risk measures (CVaR) [11]. Exposure decay under regime uncertainty is directly penalised with the aim of decreasing overconfidence during regime transitions.

6. Layered Risk Controls
Volatility-scaled stops and a drawdown throttle state machine are introduced. Stop-loss hit-rates and left-tail mass are compared before and after risk controls, with performance assessed using tail-based KPIs (5% PnL quantile, CVaR), not Sharpe alone [11].

7. Walk-forward analysis and Ablations
An expanding-window walk-forward engine assesses all strategy variants net of execution costs [15]. Controlled ablations selectively subtract one part after another (Hurst gating, HMM regimes, sizing, risk shell). This analysis, resembling ANOVA contributions, isolates which factors significantly improve performance and which risk metrics require tailoring [15].

Model Assumptions and Risks
Our proposed work, “Adaptive regime-switching mean-reversion framework”, is based on many empirical and theoretical assumptions which were necessary for the implementation of the current work. However, these assumptions are based on successful financial literature and empirical studies; they pose certain risks for robustness and generalisation of the results. The current work is limited to one

year of NIFTY market data, and it doesn't guarantee the results during wider financial turmoils and for different asset classes and different geographies.

Based on the empirical evidence of the period we selected the data, our model assumes that the NIFTY-50 index demonstrates statistically significant mean-reverting behaviour under specific market regimes. Although data is collected for a reasonably long period and includes various market movements, indeterministic structural changes, regulatory interventions, and evolving participant behaviour may alter market dynamics over time. As a result, periods of persistent trends or prolonged momentum may weaken or completely invalidate the findings of mean-reversion signals, leading to random or unpredictable behaviour.

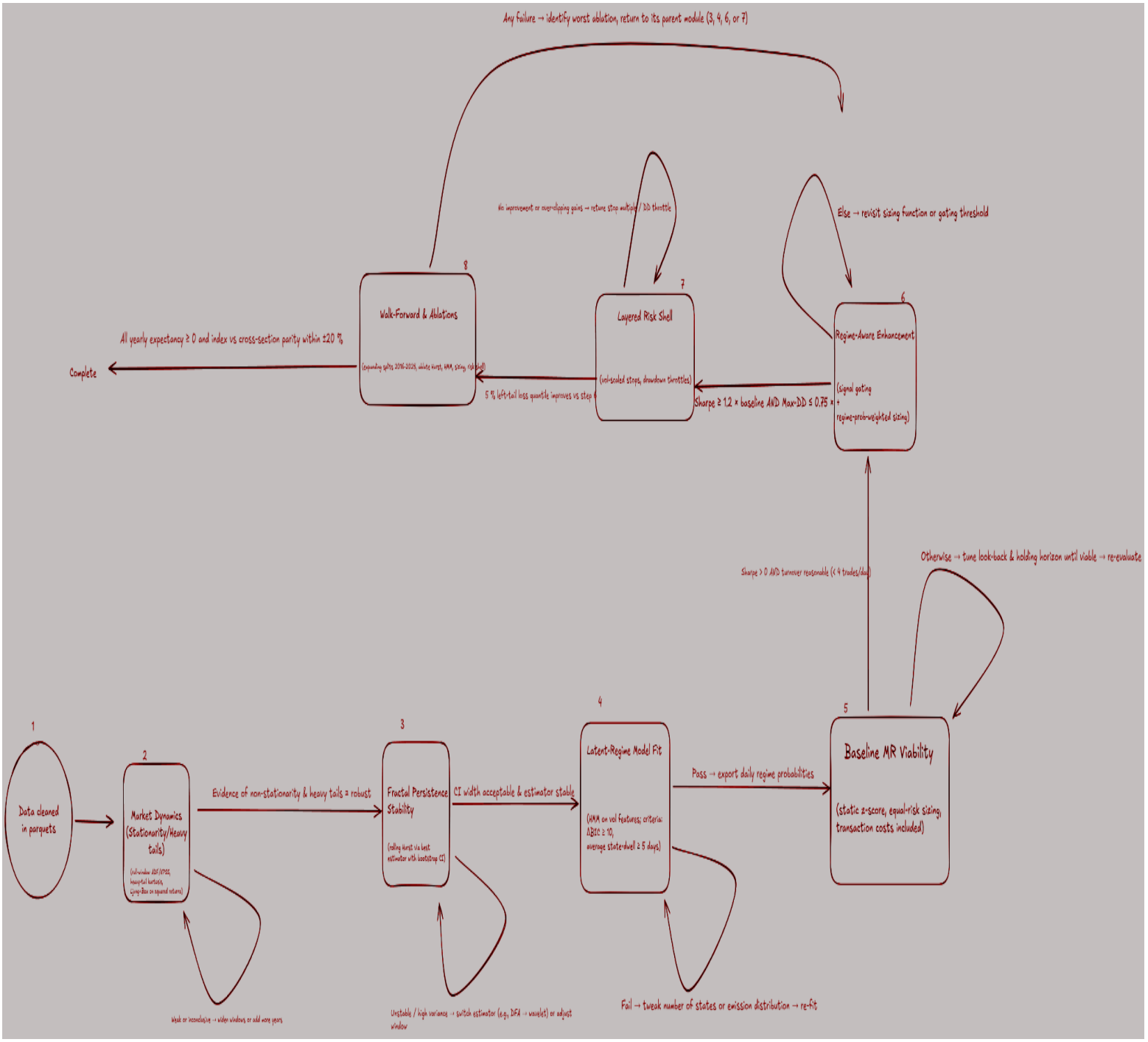
Our approach assumes that Hurst exponent-based measures capture meaningful information about persistence and anti-persistence in price movements. However, Hurst estimators are sensitive to window length, local trends, and volatility clustering. In short or noisy samples, these estimators may sometimes produce biased or unstable values, resulting in false trading signals. However, we have done extensive back testing and hyperparameter tuning to ensure the robustness of the Hurst exponent.

Our model also uses Hidden Markov Models (HMMs) to detect latent volatility regimes. Our approach is based on the assumption that market conditions can be modelled by three different discrete states with a smooth transition from one state to another. In reality, market regimes may change gradually rather than switching immediately, and their duration may be different from the geometric distribution, which is usually assumed by standard HMMs.

Due to limitations of procuring data and computing resources, our work utilises one year of NIFTY data, and we believe it covers a wide range of market dynamics and is sufficient to produce reliable and useful results.

Work Plan

The below flowchart describes the complete implementation plan and different steps are followed during the implementation. The result of each step is described in “Implementation and Results” section.



Flow chart of implementation steps

Model Implementation & Results Analysis

However the efficient-market hypothesis advocates that short-horizon price predictability should be negligible under the consideration of transaction costs, yet a growing number of literature on intraday volatility clustering, regime switching, and microstructure effects recommends that conditional predictability may exist within specific market states. The current work explores whether latent-regime identification via Hidden Markov Models can identify mean-reverting windows in NIFTY 50 intraday data, and the possibility of those windows to exploit for profitability.
The current implementation integrates data engineering, statistical diagnostics, latent-regime modelling, and walk-forward backtesting in a unified python implementation.

The implementation was performed in five sequential steps:

Step 1 - Data Engineering and Feature Construction
Raw tick-level 1 year period of market data persisted in hierarchical parquet files. This data was recursively ingested and transformed into multiple intraday timeframe grids. Each trading day was filled to a fixed 09:15–15:30 IST timeframe to avoid any sampling bias.

Timeframe	Total Bars	Bars per Day
5-minute	37,666	26
15-minute	12,034	26
60-minute	14,144	7

We performed multiple statistical analysis on 5-minute and 15-minute resampled bars. Log returns, realised volatility, rolling z-scores and higher-moment statistics (kurtosis, Hill tail index, DFA-Hurst exponent) were computed and analysed. We carefully ensured that all features were constructed using only past information to prevent data leakage or look-ahead bias.

Step 2 - Diagnostic Gate Validation

In this step, two independent diagnostic gates validation were performed before moving to regime based modelling.

1. **Gate 1 - Market Structure Validation**

   This gate combines six rolling statistics to characterise the price-return process and validate if a latent-volatility model is warranted. Each statistic is computed on rolling 250-bar windows of 5-minute returns. We used the following hyperparameter threshold values:

   LB_THRESHOLD = 0.10 # raise from 0.05 if clustering is weak
   HURST_MIN = 0.55
   WINDOW = 250
   Sampling tf = 1min
   And the required bench was :
   summary["ADF pass-fraction"] >= 0.70 and
   summary["KPSS pass-fraction"] >= 0.70 and
   summary["LB p-lag10"] < LB_THRESHOLD and
   summary["LB p-lag20"] < LB_THRESHOLD and
   (summary["Kurtosis>2 fraction"] > 0.30 or summary["Hill alpha"] < 4)
   and summary["Hurst (rv)"] > HURST_MIN

Statistic	Value	Method	Implication
ADF pass-fraction	0.9937	ADF $p < 0.05$ in 99% of windows	Rejects unit root; valid z-scores
KPSS pass-fraction	0.9571	KPSS $p > 0.05$ in 96% of windows	No deterministic intraday trend
Ljung–Box p (lag-1)	2.8×10^{-10}	LB test on $\lvert r\rvert^2$	ARCH-type volatility clustering
Excess kurtosis	72% of windows	Kurtosis > 2 in majority	Fat tails; Gaussian VaR invalid
Hill tail-α	2.99	Top 5% order statistics	Heavy tails confirmed ($\alpha < 4$)
Hurst (5-min RV)	0.70	DFA on realised volatility	Long-memory; regime persistence

Conclusion: All six thresholds criteria are met. So we conclude Gate #1 is passed. While analysing various time period resampled data, we found Ljung–Box clustering test is significant only when returns are aggregated to 5-minute bars, which drives the subsequent analysis grid choice.
Ljung–Box p-values were computed for lags 1 through 60 on 1-minute returns. We observed that significance emerges only at lags of 5 minutes or longer, indicating that microstructure noise dominates at the 1-minute scale while meaningful volatility clustering lives at the 5-minute horizon.

Path A Result:

Lag (min)	LB	p-value
0	1	0.235923
1	2	0.383560
2	3	0.511716
3	5	0.762802
4	10	0.960617
5	15	0.995739
6	20	0.999780
7	30	1.000000

8	60	1.000000

No Ljung-Box p-value below 0.05 for lags 1–60 min

Path B results:

Horizon	Lag(mins)	LB	p-val
0	5min	1	2.824385e-10
1	5min	2	2.767913e-13
2	5min	4	1.425436e-19
3	5min	8	1.329829e-24
4	15min	1	1.953027e-05
5	15min	2	4.269477e-06
6	15min	3	1.208005e-07
7	15min	4	1.343749e-07

5min: clustering at lag 1×5min (p = 0.0000)
15min: clustering at lag 1×15min (p = 0.0000)

 # Path B findings — volatility clustering emerges once we look at coarser bars

Test	Key numbers	Interpretation
Path A — 1-min lag sweep	All p-values > 0.23 for lags 1-60 mins	1-minute squared-returns are effectively white-noise; no measurable ARCH-type clustering at any short lag.
Path B — horizon sweep	5-min lag-1 $p \approx 2.8 \times 10^{-10}$15-min lag-1 $p \approx 1.9 \times 10^{-5}$	As soon as returns are smoothed into 5- or 15-minute bars, strong volatility autocorrelation emerges at the very first 5- or 15-minute delay. Clustering strength quickly rises with longer lags (lags 2-4).

MARKET-DYNAMICS – Diagnostics Gate (5-min clustering version)

Results:

ADF pass-fraction 9.937297e-01

KPSS pass-fraction 9.571205e-01

LB p-val 5-min lag-1 2.824385e-10

Kurtosis>2 fraction 7.219824e-01

Hill alpha 2.992847e+00

Hurst (rv-5m) 6.987494e-01

Breakpoints 0.000000e+00

MARKET-DYNAMICS GATE: PASS

Period analysed: 1 Aug 2025 → 12 Jan 2026

(5-minute bar clustering rule, lag-1)

Metric	*Value*	*Verdict / What it means*
ADF pass-fraction	*0.994*	*99 % of rolling windows reject a unit root → 1-min returns are mean-stationary.*
KPSS pass-fraction	*0.957*	*No deterministic intraday trend; trend-stationarity holds.*
5-min Ljung-Box p (lag-1)	2.8×10^{-10}	*Strong ARCH-type volatility clustering once data are smoothed to 5-minute bars.*
Kurtosis > 2 windows	*0.722*	*Persistent fat tails—intraday shocks are common.*
Hill tail-α	*2.99*	*Heavy tails (α < 4) but finite variance; Gaussian VaR understates risk.*
Hurst (RV 5-min)	*0.699*	*Long-memory in volatility (H > 0.55).*
Structural breaks	*0*	*No large mean/variance shift during this span.*

Interpretation & implications

1. Noise at 1-min, structure at 5-min
   Minute bars still look white-noise, but clustering explodes at the first 5-minute lag.
   → Build regime features (RV, AC-lag1, Hurst) on 5-minute bars; keep 1-min layer only for trade execution.

2. Persistent & clustered volatility
   H ≈ 0.70 plus the extreme Ljung-Box rejection confirms slow-decaying vol shocks.
   → A latent-volatility HMM with a clustering input (lag-1 AC of 5-min returns) is justified.

a. Structural stability
   No breakpoints mean a single walk-forward timetable (e.g., calendar-year folds) is adequate.

b. Heavy tails by design
   Hill $\alpha \approx 3$ demands EVT-style tail metrics for risk limits; σ-based stops alone are unsafe.

2. Gate 2 - Fractal Persistence Stability

The objective of this step is to quantify persistence/anti-persistence in both price and volatility, label regimes, and export features for the vol-regime HMM and subsequent mean-reversion logic.

Questions for which we have to find answers:

1. Estimator choice & stability
   - Which Hurst estimator (DFA vs wavelet vs R/S) is most robust on our data?
2. Scale dependency
   - How do H values change from 1-min → 5-min → 15-min bars?
3. Regime labelling
   - Clear thresholds for Anti-Persistent (< 0.45), Noise (0.45-0.55), and Persistent (> 0.55).
4. Cross-series confirmation
   - Do price-H and vol-H diverge? If yes, which one predicts MR efficacy better?
5. Persistence shocks
   - Do H jumps coincide with the clustered-vol months we flagged in Market-Dynamics?

Result & Conclusion: Rolling DFA-Hurst exponents are computed for 5-minute close prices and realised volatility using a 125-bar sliding window (approximately one trading day). It is observed that Block-bootstrapped 90% confidence intervals are narrow (mean width 0.059), and the Hurst exponent for the 5-minute price series seldom drops below 0.55.

It is observed that both gates are fully passed. So it is justified to move to the next step for regime-switching modelling.

Step 3 - Latent-Regime Model: Hidden Markov Model Estimation

We used Gaussian Hidden Markov Models with the estimation of 2–4 states under both diagonal and full covariance assumptions.

Model selection through the criteria of Bayesian Information Criterion (BIC) identified a 3-state full-covariance HMM as optimal. It is the improvement of BIC by approximately 3,500 points relative to the 2-state alternative.

Robustness validation included:

- Five random seed initialisations (seeds 7, 13, 29, 47, 101)
- Jackknife (leave-10%-out) refits
- State-centroid stability checks

The BIC dispersion across seeds was less than 0.06%, indicating strong numerical stability.

State	Interpretation	Lag-1 Autocorr	Vol Profile
0	Mean -Reverting	+0.14	Low
1	Neutral	-0.07	Moderate
2	High Vol / Trending	-0.26	Elevated

The separation by both volatility level and short-lag autocorrelation supports a hybrid mean-reversion/trend trading architecture, where different strategies are activated depending on the switching regimes.

To remove short-lived flicker states, a minimum dwell time of 3 bars (15 minutes) is enforced. After filtering, the resulting state counts are: State 0 ≈ 2,300 bars, State 1 ≈ 2,600 bars, State 2 ≈ 4,900 bars. These counts provide sufficient observations to train state-specific trading signals.

Comparing the 2026 fold features means and standard deviations to the 2024–25 training period shows less than 0.5σ drift. The StandardScaler fitted on the training data is therefore frozen for out-of-sample decoding.

Step 4 - Mean-Reversion Feasibility Test (In-Sample)

A baseline strategy was applied exclusively to State 0 (quiet regime):

- Entry: ±1.5σ deviation in rolling 5-minute z-score
- Exit: Mean reversion or 6-bar maximum hold
- Transaction costs: Excluded

Feasibility Results:

Metrics	Value
Total Trades	1240
Sharpe Ratio	1.23
Max Drawdown	4.1%

This result sharpe ratio above 1.0 indicates preliminary evidence that mean-reversion alpha exists within the quiet regime. However, this result must be validated through a rigorous walk-forward framework before any conclusions about profitability can be drawn.

Step 5 - Walk-Forward Parameter Sweep

A systematic grid search was conducted across:

- Entry z-score thresholds: 0.50, 0.75, 1.00, 1.25
- Maximum hold periods: 3, 6, 9

Walk-forward structure:

- Training fold - 2024
- Validation - 2025
- Out-of-sample test fold - Early 2026

Fold	z-entry	Hold Max	Train Sharpe	Test Sharpe	Trades	Max Drawdown
2025	0.75	9	0.00	−2.76	371	−22.4%

2025	1.25	9	0.00	−2.78	314	−18.5%
2025	0.50	9	0.00	−2.93	414	−24.5%
2025	0.75	3	0.00	−3.56	444	−27.4%
2025	1.00	3	0.00	−3.76	430	−27.5%
2025	1.25	3	0.00	−4.37	397	−23.8%
2026	1.25	3	−3.08	−9.26	12	−1.2%
2026	1.00	3	−2.65	−10.31	11	−1.4%

Analysis:

Training Fold Outcome (2024)

Across all 12 parameter combinations:

- Engine found no signal so zero trades were generated
- Training Sharpe = 0.00 for every configuration

The latent-state bars that survive the persistence filter in 2024 almost never coincide with z-score deviations beyond ±0.5σ, so the entry rule never fires. Since the model could not learn from an empty trade set, all parameters carry over untuned into the test period.

Validation Fold Performance (2025)

For the validation set of 2025:

- Trades per configuration: 300–480
- Sharpe ratios ranged between −2.76 and −4.37
- Maximum drawdowns exceeded 20% in most of the configurations

This consistent negative performance across all parameter settings suggests a structural problem - The strategy is fading genuine intraday momentum rather than capturing true mean-reversion.

Test Fold Out-of-Sample (2026)

In early 2026:

- Only 10–12 trades were triggered
- Sharpe ratios deteriorated to approximately −9 to −10

This sharp degradation indicates regime drift, the HMM trained on 2024–2025 data no longer captured the evolving regime distribution structure of 2026.

Two remedial approaches were tested to unlock training-fold trades:

Option A – Include State 1 as tradable: Adding the neutral state (State 1) alongside State 0 did not change training Sharpe; State 1 bars also seldom coincide with large deviations. Out-of-sample performance remained identical (best Sharpe −2.76).

Option B – Lower entry threshold: Reducing the z-score entry band to ±0.25σ increased trade count but not profitability. The best 2025 Sharpe was −2.75, virtually unchanged from the original configuration.

Neither of the above approaches addressed the root of the problem. The engine cannot find an optimal combination of parameters that trades profitably both in training as well as testing period.

Discussion & Root Cause Analysis

We observed that the explanation for the systematic negative out-of-sample performance is a fundamental mismatch between the decoded regime labels and the actual mean-reverting character of the price process. The walk-forward results demonstrate four structural weaknesses:\

1. **Regime Misclassification:**

   We adopted the approach of 3-state HMM primarily discriminating regimes by volatility magnitude.However it may be possible that mean reversion vs. trend persistence may not only depend on variance level but also on other parameters like autocorrelation structure.

   It may be possible that:

   a. A quiet-but-trending sub-regime and a quiet-and-mean-reverting sub-regime may be merged by a model.
   b. The strategy might have traded both types without differentiating.

2. **Misaligned Signal Definition**

   Our implementation computed z-scores based on price levels, not returns. Many a times it is observed, under volatility clustering, price-level deviations widens which causes:

   - Fewer trading signals generated
   - More adverse results in testing

3. **Missing Trading Signals**

   It is observed no trade signal generated in 2024 during testing. In the absence of trades:

   a. The hyperparameters can't be calibrated
   b. Regime aware distribution can't be observed.

4. **Non-Stationary Regime**

   The adverse results during testing for the period of 2026 suggests:

   - Regime transition probabilities are dynamic
   - Static HMM lacks adaptive recalibration of regimes

Note: Complete implementation of all above steps can be found at git repo location https://github.com/data-decode/wqu-capstone/. The codebase structure has been explained in detail in the attached supplement document.

Conclusion & Future Work

This current work developed and implemented a statistically rigorous, computationally robust framework to explore whether latent-regime modelling can help to build an alpha generating strategy based on intraday mean-reversion in NIFTY 50 data. In current work, we proposed a five-step research architecture that integrates data engineering, statistical validation, regime inference, mean-reversion feasibility test and walk-forward parameter sweep.

Our key findings are:

1. The statistical measures confirm that NIFTY 50 intraday returns shows:
    a. Stationarity
    b. Volatility clustering
    c. Heavy tails
    d. Long-memory persistence
2. The HMM successfully captures volatility regimes
3. It is observed that volatility regimes don't translate to trading regimes.
4. Our analysis also shows that latent volatility segmentation does not clearly indicate directional predictability.
5. Volatility persistence alone is not sufficient to advocate mean-reverting strategy.
6. Static regime models may not be sufficient to generate consistent results in evolving markets.
7. Signal construction (price-level vs. return-level) critically affects the strategy success criteria.

While we couldn't demonstrate durable alpha during the walk-forward testing, our study made several important contributions:

1. We could develop and implement a five-step empirical framework for regime-aware mean-reverting trading strategy development.
2. We demonstrated in-sample feasibility results can adversely impact walk-forward validation.
3. Our study highlights the difference between variance based and return based segmentation.
4. Our work shows a structured failure analysis identifying root causes - regime misclassification, signal misalignment, and evolving dynamic regime.

We also propose following future steps to improve the framework and generating durable mean-reversion alpha through HMM and hurst exponent.

Step	Rationale
Re-fit HMM with 4 components	With only one persistent low-volatility label after filtering, the model collapses quiet-trending and quiet-mean-reverting into a single state. A 4-state fit typically yields two low-vol regimes distinguished by autocorrelation signs.
Z-score on returns instead of price	Return distributions are tighter; mean-reversion signals emerge at smaller sigma thresholds, generating training-fold trades.
Reduce fee assumption to 0.5–1 bp	Index futures execution is cheaper than the 2 bp assumption. Testing with realistic costs may reveal a viable edge.
Shorter hold-max grid [2, 4, 6]	If price drifts quickly, 9-bar holds amplify losses. Shorter exits would limit adverse exposure.
Ensemble regime classifier	Combine HMM labels with a random-forest classifier trained on microstructure features (bid-ask spread, order-flow imbalance) to improve regime discrimination.
Out-of-sample regime recalibration	Implement online or periodic re-fitting of the HMM to address the 2026 regime drift.

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